Cluster 55922

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Subjects

Research fields: Computer science, Engineering, Mathematics

The most common research fields of articles in this cluster. Read more.

Research subfields: Reliability engineering, Operations research, Mathematical optimization

Common research subfields of articles in this cluster. Read more.

Key concepts: Demand Response, power grid, power system, power consumption, energy storage

These key concepts are identified algorithmically using article titles and abstracts. Read more.

1.96% of all articles are AI-related

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. Read more.

0.28% of English-language articles are related to robotics

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to computer vision

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to natural language processing

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to AI safety

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.28% of English-language articles are related to cybersecurity

Vital signs

357 recent articles in the cluster

357 articles in this cluster were published in the five year period ending in 2023.

Average article is **8.29** years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **27.87th** percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 27.87% of other clusters.

Citation rating: **17.38th** percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year.In this case, the average article in this cluster is cited more often than roughly 17.38% of all other articles published in the same year.

51.54% of articles are in English

Not expected to grow extremely fast

As determined by ETO's growth prediction model. Read more.

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 348 of 357 articles. 348 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
China	338	0.27
India	4	0.06
United States	3	0.67
South Korea	1	0.40
Singapore	1	0.40
Sri Lanka	1	0.25
Russia	1	1.00
Denmark	1	1.00

Top funding countries

Data available for 41 of 357 articles. 41 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
China	41	0.76

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

Demand response strategy for electric vehicles and air conditioners considering response willingness
2023: 电力系统保护与控制. 0 citations.

Modelling and Simulation Study of TOU Stackelberg Game Based on Demand Response
2020: 电网技术. 0 citations.

A summary of time-of-use research and practice in a demand response environment
2021: 电力系统保护与控制. 0 citations.

Distributionally Robust Modeling of Demand Response and Its Large-scale Potential Deduction Method
2022: 电力系统自动化. 0 citations.

Design of Demand-Response Market Mechanism in Accordance with China Power Market
2021: 电力建设. 4 citations.

Key Technologies and Prospects of Demand-side Resource Utilization for Power Systems Dominated by Renewable Energy
2021: 电力系统自动化. 0 citations.

Multi-objective optimal operation of distributed energy storage considering user demand response under time-of-use price
2020: 电力自动化设备. 0 citations.

Source-load-storage coordinated optimal scheduling of microgrid considering differential demand response
2020: 电力自动化设备. 0 citations.

An incentive strategy of residential peak-valley price based on price elasticity matrix of demand
2021: 电力系统保护与控制. 0 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

No data available.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

A Comprehensive Clustering Method of User Load Characteristics and Adjustable Potential Based on Power Big Data
2021: 中国电机工程学报. 27 citations.

Modelling and Simulation Study of TOU Stackelberg Game Based on Demand Response
2020: 电网技术. 0 citations.

5G Communication Base Stations Participating in Demand Response:Key Technologies and Prospects
2021: 中国电机工程学报. 42 citations.

Key Technologies and Prospects of Demand-side Resource Utilization for Power Systems Dominated by Renewable Energy
2021: 电力系统自动化. 0 citations.

Multi-objective optimal operation of distributed energy storage considering user demand response under time-of-use price
2020: 电力自动化设备. 0 citations.

Source-load-storage coordinated optimal scheduling of microgrid considering differential demand response
2020: 电力自动化设备. 0 citations.

Economic optimization model of a load aggregator based on the multi-agent Stackelberg game
2022: 电力系统保护与控制. 28 citations.

An incentive strategy of residential peak-valley price based on price elasticity matrix of demand
2021: 电力系统保护与控制. 0 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 357 out of 357 articles. 136 of these articles came from these sources.

Source	Articles (last five years)
电力系统自动化	24
电网技术	18
电力自动化设备	18
中国电力	14
电力系统保护与控制	14
电力建设	12
Frontiers in Energy Research	11
南方电网技术	9
Journal of Physics: Conference Series	9
电工技术学报	7

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 356 of 357 articles. 27 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
李彬	10	0.00
Li Bin	10	0.00
Qi Bing	9	0.00
祁兵	9	0.00
孙毅	8	0.00
Sun Yi	8	0.00
Yi Wang	5	2.27
奚培锋	4	0.00
Xi Peifeng	4	0.00
Songsong Chen	4	0.88

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 349 of 357 articles. 113 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
North China Electric Power University - China	26	0.64
State Grid Corporation of China (China)	23	0.45
(北京)华北电力大学电气与电子工程学院 - China	15	0
浙江大学电气工程学院 - China	11	0
China Electric Power Research Institute - China	10	0
Zhejiang University - China	9	0.6
China Southern Power Grid (China)	9	0.79
Southeast University - China	9	0.26
上海电力大学电气工程学院 - China	8	0.03
Tianjin University - China	8	0.51

Organization types

This table covers the 199 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 357 articles total). Read more

Commercial	Education	Nonprofit	Government
12.66%	35.23%	8.86%	0.63%

Top industry organizations

The industry organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 59 out of 357 articles. 53 of these articles had at least one author from an organization listed below.

Organization	Articles (last five years)	Yearly citations (average)
State Grid Corporation of China (China)	23	0.45
China Southern Power Grid (China)	9	0.79
Shanghai Electric (China)	7	0.55
State Grid Hebei Electric Power Company - China	5	0
State Grid Nanjing Power Supply Company (China)	3	0
Jilin Electric Power Research Institute (China)	2	0
Tianjin Research Institute of Electric Science (China)	2	0
Henan Energy & Chemical Industry Group (China)	1	0
China Design Group (China)	1	1
China Power Engineering Consulting Group (China)	1	0

Top funders

Data available for 78 out of 357 recent articles. 40 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Natural Science Foundation of China	14	0.32
State Grid Corporation of China (China)	11	0.39
Ministry of Science and Technology - China	5	0.40
Ministry of Education - China	3	1.25
Ministry of Civil Affairs - China	3	0.82
State Grid Corporation of China	3	0.94
Science and Technology Project of NARI Technology Co., Ltd on "Research on Key Technologies for Simulation and Evaluation of Comprehensive Energy Systems"	3	0.20
National Key Research & Development Program of China	3	0.67
Science and Technology Project of State Grid - China	3	0.94
Science and technology projects from State Grid Corporation	3	0.00

Funder types

This table covers the 30 articles in this cluster from the last five years with data about their funding organization types (out of 357 articles total).

Commercial	Education	Nonprofit	Government
38.24%	2.94%	5.88%	52.94%

Patents and industry

58 patents cite articles in this cluster (**70.86th** percentile among all clusters)

5.1% of articles in the cluster are cited by patents

8.12% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
TIME-OF-USE ELECTRICITY PRICE PRICING METHOD SUITABLE FOR PARK COMPREHENSIVE ENERGY SUPPLY SERVICE	1
HOUSEHOLD LOAD SCHEDULING METHOD	1
POWER TRANSMISSION AND DISTRIBUTION NETWORK COLLABORATIVE OPTIMIZATION CONTROL METHOD BASED ON MASTER-SLAVE GAME	1
DISTRIBUTED PHOTOVOLTAIC-BASED INTELLIGENT CLUSTER CONTROL METHOD AND SYSTEM	1
CAPACITY OPTIMIZATION CONFIGURATION METHOD OF INDEPENDENT MICRO POWER GRID SYSTEM	1
ORDERLY POWER UTILIZATION CONTROL METHOD AND SYSTEM IN POWER UTILIZATION PEAK PERIOD	1
POWER GRID BLOCKING RISK ASSESSMENT METHOD AND SYSTEM CONSIDERING NODE CONNECTIVITY LOSS	1
SOLAR ENERGY TIME-SHARING PRICING METHOD AND SYSTEM BASED ON CONDITION CREDIBILITY	1
POWER DISTRIBUTION NETWORK FLEXIBILITY EVALUATION METHOD AND DEVICE CONSIDERING ADJUSTABLE CAPABILITY OF FLEXIBLE RESOURCES	1

Top patent assignees

Patent assignee	Same-cluster citations
STATE GRID CORPORATION OF CHINA (CHINA)	6
UNIV NINGXIA	3
国网浙江省电力有限公司	3
国网河北省电力有限公司营销服务中心	2
ZHEJIANG PROVINCE ELECTRIC POWER LTD COMPANY OF STATE GRID	2
MARKETING SERVICE CENTER OF STATE GRID HEBEI ELECTRIC POWER CO LTD	2
HUAWEI TECHNOLOGIES (CHINA)	2
NORTH CHINA ELECTRIC POWER UNIVERSITY	2
STATE FARM (UNITED STATES)	16

Connections

Top citing clusters

Cluster	Number of significant citations	CSET extracted phrases
837	3	home energy management, energy management system, Smart Home Energy, smart grid, renewable energy
18540	2	smart meter data, energy consumption data, electricity consumption, load profiles, Electric Load Clustering
4403	2	Wind Power, power system, Wind Power Uncertainty, renewable energy, robust optimization
6604	1	energy storage system, power system, thermostatically controlled loads, demand response, Frequency Regulation
27535	1	Virtual power plant, distributed energy resources, VPP energy management, renewable energy, multi-energy virtual power
4332	1	demand response, demand side management, incentive-based demand response, electricity demand, power system

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
837	22	home energy management, energy management system, Smart Home Energy, smart grid, renewable energy
2535	17	integrated energy system, Energy Systems, Park-level Integrated Energy, demand response, renewable energy
6604	15	energy storage system, power system, thermostatically controlled loads, demand response, Frequency Regulation
20	14	Electric Vehicle Charging, Vehicle Charging Load, charging electric vehicles, charging stations, EVs
4332	11	demand response, demand side management, incentive-based demand response, electricity demand, power system
4403	8	Wind Power, power system, Wind Power Uncertainty, renewable energy, robust optimization
2628	3	energy storage system, battery energy storage, BESS, wind power, renewable energy
21840	3	Energy consumption, energy-saving behaviors, planned behavior, household energy, consumption behavior
18540	3	smart meter data, energy consumption data, electricity consumption, load profiles, Electric Load Clustering
11721	2	Power System, Commutation Failure, HVDC Transmission System, multi-infeed HVDC systems, HVDC Systems

Top GitHub repositories

No data available.